

Experiment 6

Implementation of Diffie–Hellman Key Exchange Algorithm

Aim

To implement the Diffie–Hellman Key Exchange algorithm for securely generating a shared secret key between two users.

Description

The Diffie–Hellman Key Exchange algorithm is a public-key cryptographic technique that allows two parties to establish a common secret key over an insecure communication channel. Both users generate their public keys using a common prime number and primitive root. After exchanging public keys, each user computes the same shared secret key independently without transmitting it.

Algorithm

1. Start the program.
2. Enter the prime number p .
3. Enter the primitive root g .
4. Enter the private keys of User A and User B.
5. Compute the public keys.
6. Exchange the public keys.
7. Compute the shared secret key at both users.
8. Display the public keys and shared secret keys.
9. Stop the program.

Code

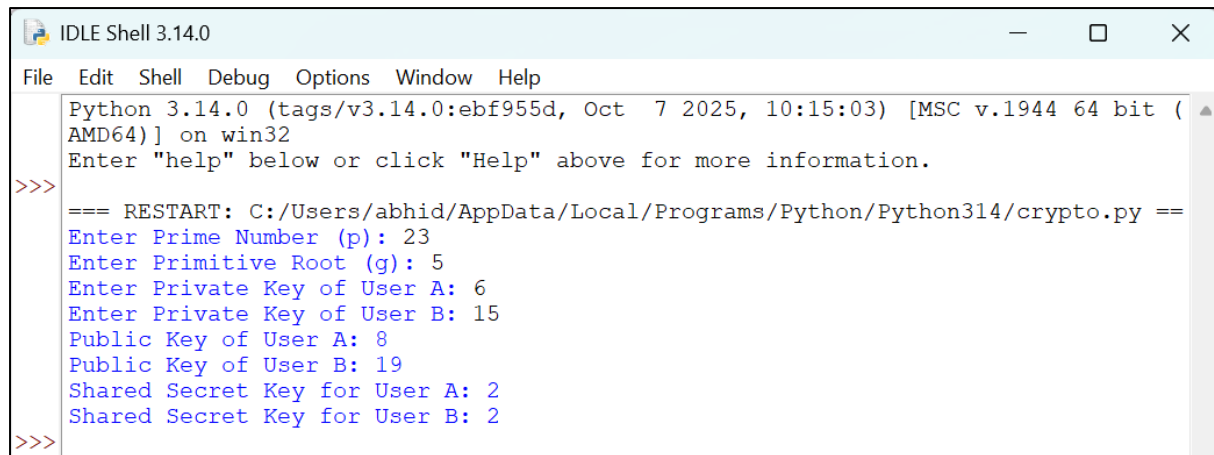
```
p = int(input("Enter Prime Number (p): "))
g = int(input("Enter Primitive Root (g): "))
a = int(input("Enter Private Key of User A: "))
b = int(input("Enter Private Key of User B: "))
A = pow(g, a, p)
B = pow(g, b, p)
keyA = pow(B, a, p)
keyB = pow(A, b, p)
print("Public Key of User A:", A)
```

```
print("Public Key of User B:", B)

print("Shared Secret Key for User A:", keyA)

print("Shared Secret Key for User B:", keyB)
```

Output:



The screenshot shows a window titled "IDLE Shell 3.14.0" with a menu bar (File, Edit, Shell, Debug, Options, Window, Help). The shell displays the following output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/abhid/AppData/Local/Programs/Python/Python314/crypto.py ===
Enter Prime Number (p): 23
Enter Primitive Root (g): 5
Enter Private Key of User A: 6
Enter Private Key of User B: 15
Public Key of User A: 8
Public Key of User B: 19
Shared Secret Key for User A: 2
Shared Secret Key for User B: 2
>>>
```